



Bayesian Spam Filters

Analyse, learn, evolve

The issue of spam has been getting bigger over the years, with more legislation, more discussion, and more solutions. Suggestions range from strict anti-spam legislation, to Bill Gates' penny-a-mail proposal. Whatever turn the issue may take next, it's certain that spam filters will play a role in the much-talked-about battle. And in spam filtering, Bayesian seems to be the way to go.

Spam filtering is ideally suited to Bayesian methods in general, which use past knowledge to predict future events. Bayesian spam filters are radically different from other filters in several ways.

An example of the this would be a pure keyword-based filter, which only looks at the occurrences of words in e-mails to determine if a sample is spam or not. In contrast, Bayesian spam filters see causal connections, amongst other things. Thus, for example, even if the word 'debt' is evidence of spam, and the phrase 'get out of' is not, a Bayesian filter would learn from experience, without human assistance, that the phrase 'get out of debt' is indeed indication of spam.

Another plus is that Bayesian filters evolve with spam, which means they won't be outdated even as spammers devise newer techniques. And they are robust—they learn both from positive examples, as well as negative ones. Thus, the fact that 'Bayesian' hardly occurs in spam, which is a 'positive' example, makes as much difference as the fact that 'debt' frequently does.

An example of a simple Bayesian filter is a Naïve Bayes Classifier (NBC). An NBC takes just two factors into account while filtering spam. The first is the number of spam e-mails you receive. The second is

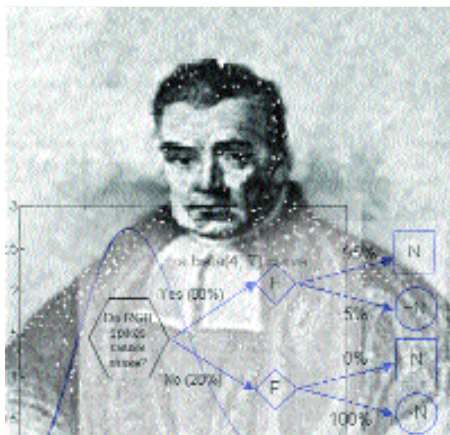
the ratio of probabilities of a given word being a 'spam' word to it not being one. For example, if the word 'debt' occurs in spam with an 80 per cent chance, and in genuine e-mail with a 2 per cent chance, this second factor would be 40.

Surprisingly, NBCs are pretty effective, even though their working is as simple as this! PopFile, an open-source spam filter, uses an NBC. There are Bayesian filters that use more advanced techniques, and also combinations of techniques, but they're all based on the same premise—that of learning from the past, that is, learning from spam and non-spam e-mails that have been analysed earlier.

On one level, it may seem intuitive that Bayesian methods would be the most effective against spam: we learn to delete spam quickly from our inboxes only because we've seen such e-mails in the past. We learn from the subject

lines, which are typical, and the senders' names, which are not. And Bayesian methods do something of the sort—learning from past experience. Of course, they analyse the entire message, not just the subject lines and senders' names, so they have a lot more to draw upon. As a matter of fact, certain Bayesian spam filters are actually better than humans at detecting spam.

From many indicators, it's clear that Bayesian spam filters are here to stay. And probably the best testimony to this is the fact that Paul Graham, designer of the Arc language, and a major crusader in the battle against spam, has this to say: "Of all the approaches to fighting spam, from software to laws, I believe Bayesian filtering will be the single most effective."



The Reverend Thomas Bayes: the Church minister behind Bayes' Theorem



WAITING IN THE WINGS

Longhorn

The much-hyped next generation of Windows will come with a hefty hardware requirements tag, but offers several improvements. Better access to information with the search-focused WinFS, and a central Contacts service that recognises the primary role of PCs as communication tools are the most significant.

A prettier interface based on 3D technology and organised around Tasks is likely to be the most visible update.

Longhorn promises to change our idea of how useful an operating system can be, but by the time it debuts in 2006, it's innovations may already be available elsewhere.